The first command installs all the required modules, and the second command starts the server on your localhost, launches the app.js, and starts listening for inputs from the port (3000) specified in the index.js file.

> **NOTE** As an alternative to the steps discussed above, you may use the ready-made modules for ballot-contract and ballot-app in this chapter's code (Ballot-Dapp), following the instructions given.

4.4.3 Install MetaMask wallet

You have one more step--installing MetaMask--to complete before you begin testing the Dapp. The account addresses are needed to identify the decentralized participants. Also, transactions have to be digitally signed and confirmed by the participant (sender of Txs). The balance of the account has to be verified to ensure that it has sufficient ether (gas points) to pay for the cost of execution of functions.

You need a mechanism for accomplishing all these important operations. For this purpose, you'll use a convenient browser plugin called MetaMask (https://metamask .io) that connects web clients to the web server and blockchain provider through the RPC port, as shown in figure 4.8. MetaMask is described as a crypto wallet and a gateway to blockchain applications; it is an intelligent digital wallet, and it acts as a proxy between the web client and the blockchain server. MetaMask securely manages the accounts created on the blockchain and their balances in ether, and digitally signs the transactions issued by the participant accounts.